

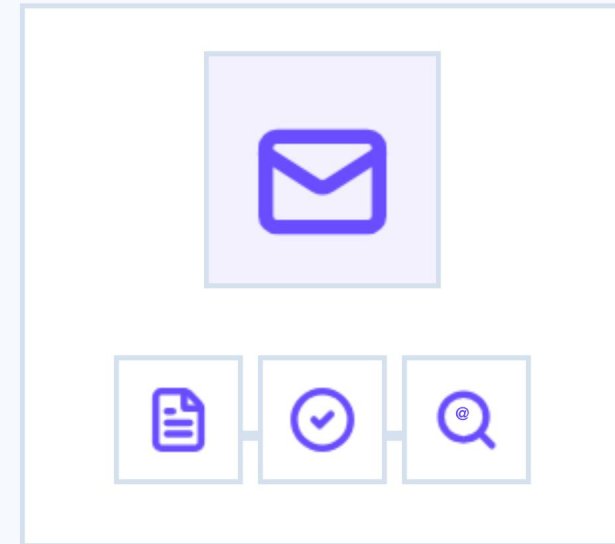
Cloud Email Analyzer

Serverless Email Threat Detection with AWS, ML, and ClamAV

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Email evidence is fragmented before review.



Raw messages are messy

MIME variants, HTML/plain text, envelope recipients, URLs, and attachments all carry evidence.



Signals arrive separately

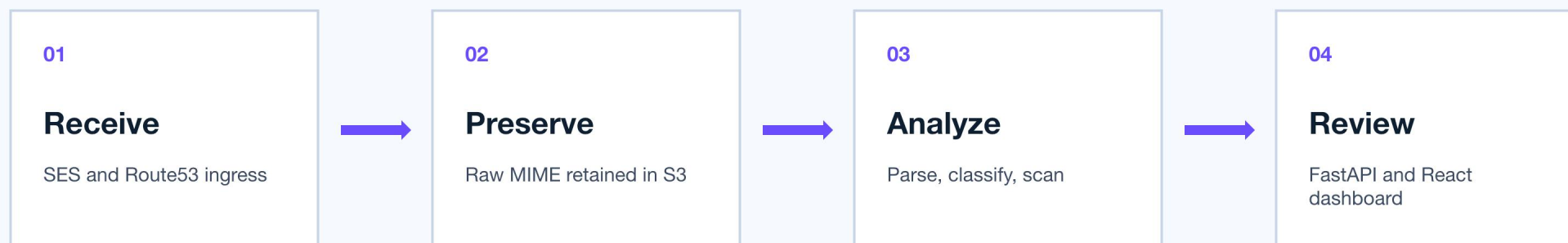
Classifier output, antivirus results, and mail verdicts must keep their source.



Review needs accountability

Scoped access, supporting evidence, and audit records are required for sensitive actions.

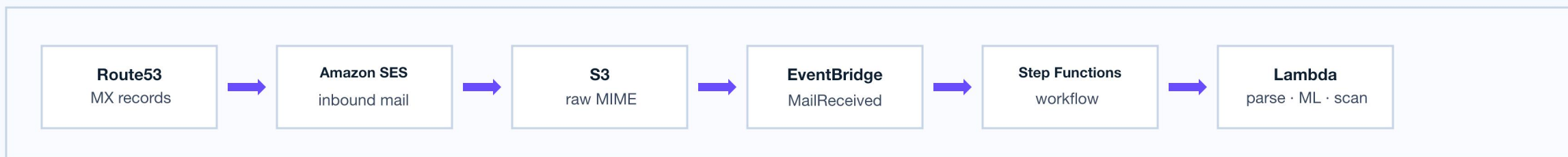
Normalize once, then analyze and review with context.



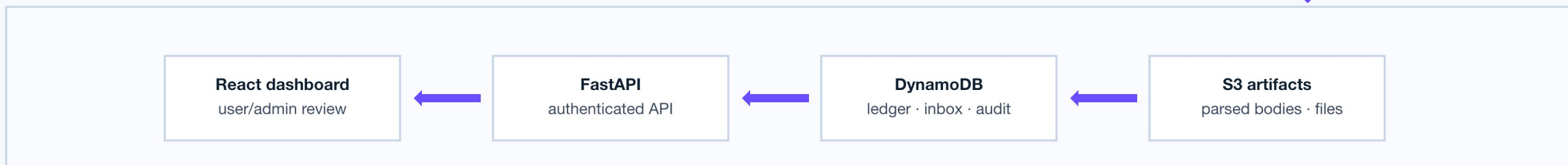
Canonical MailReceived event carries metadata and storage pointers, not raw email bodies.

AWS services separate ingestion, processing, state, and review.

Mail ingress and event boundary

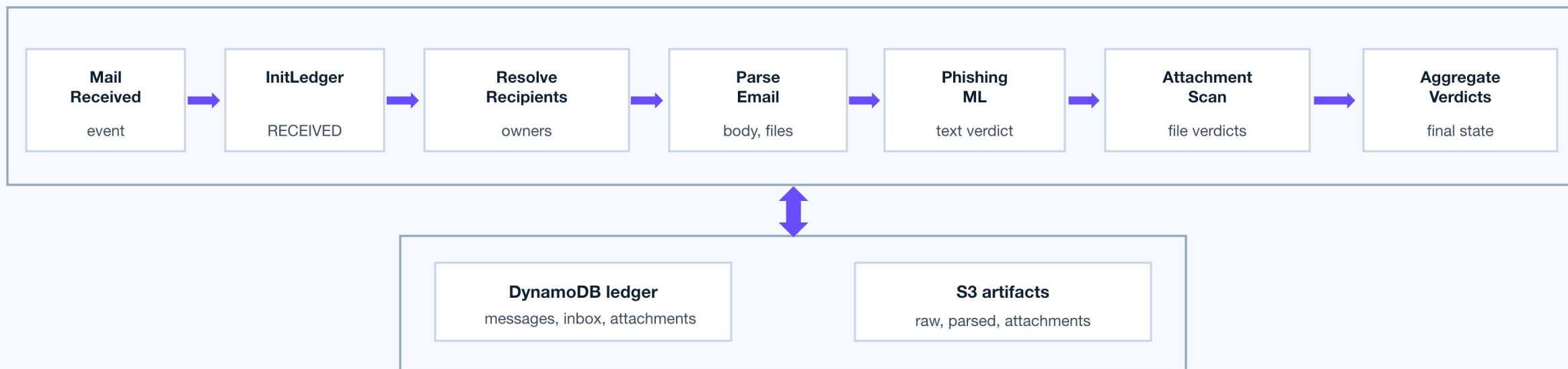


Persistence and authenticated review



**Raw bodies and extracted files remain in S3.
Compact review state remains queryable in DynamoDB.**

A normalized event moves through a short processing chain.



Processing transforms messages.
Persistence keeps evidence and review state.

The built system maps responsibilities to small modules.

Cloud services

- Route53 + SES
- S3 object storage
- EventBridge
- Step Functions
- DynamoDB

Analysis stages

- InitLedger
- ResolveRecipients
- ParseEmail
- PhishingML
- AttachmentScan
- AggregateVerdicts

Review layer

- FastAPI routes
- HS256 JWT authorization
- React dashboard
- Admin review
- Audit records

Separate artifacts from review state.

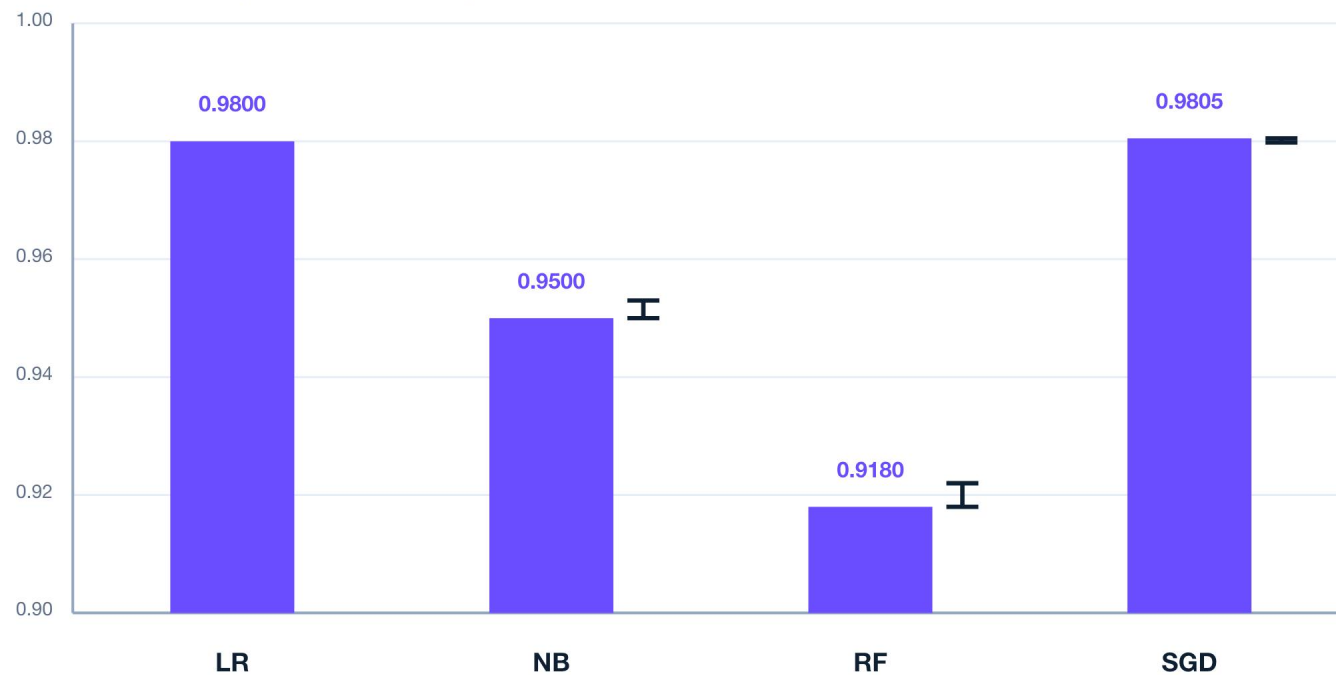


Dashboard requests compact state first, then retrieves large artifacts through authorized API calls.

Internal validation supports the packaged classifier.

Classifier recall

Bars show recall; whiskers span the tracked metric spread.



0.9805

best-model recall

Confusion matrix

TN 7,748

FP 158

FN 152

TP 7,649

The architecture leaves room for stronger evidence sources.

Current scope

- Detector scope: text classifier and ClamAV file scan.
- Policy choices: retention, quarantine, escalation, risky HTML handling.

Future work

- Add URL/domain reputation and header-authentication scoring.
- Move identity management to Cognito and manage ClamAV signatures.

Cloud Email Analyzer turns inbound mail into traceable security evidence.

- AWS-oriented event boundary
- ML classification plus attachment scanning
- Authenticated user/admin review with audit records

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